



HUD Information Design for IoV Intelligent Navigation System

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Abstract. The car navigation function is mainly realized through the smartphone and the central control interface, both of which are not in the driver's best field of vision, bringing potential safety hazards. Head-Up Display (HUD) projects information to the upper part of the ambitious vision, such as the car's front windshield, which has a better driving experience and safety. This article adopts the case analysis method to analyze the classic design cases of navigation head-up display and proposes the information organization strategy and visual design strategy of HUD. It combines design examples to verify the feasibility of the information organization strategy and design strategy and provides a theoretical basis for the car interaction design.

Keywords: HUD · Information organization · Visual design · Navigation system

1 Introduction

The Internet of Vehicles (IoV) uses sensors, networks and control technologies to make drivers, vehicles, environment, roads, roadside facilities and pedestrians an organic whole, providing car navigation systems with richer types and quantities of information, and improving the accuracy of the information, It also provides a channel for users to publish navigation-related information [1]. At present, the car navigation of the IoV includes the following features: Heterogeneous information content, which comes from the Internet, infrastructure, and mobile devices, has a wide variety of content. In addition to static traffic information such as road names, it also includes dynamic road condition information and even event information [2]. The form of information includes text, graphics, sound, animation and even video information. The multi-state interaction mode, diversified service terminal equipment provides a multi-state interaction mode for the service. In addition to physical buttons, knobs, touch screens, voice and somatosensory, interactions can also be achieved with wearable smart devices such as Google Glass and Apple Watch. The development of augmented reality and head-up displays also provide convenience and safety for services a specific guarantee.

Services are interactive, including “human-computer interaction” and reflects the characteristics of “human-human interaction”. The user's role in the service process is no longer just the receiver of information, but also the producer and creator of information.

Therefore, the navigation system is a critical information carrier for the driver to interact with the car's human-machine system. At present, the navigation function is mainly realized through the smartphone and the central control interface, but both are not in the driver's best field of view, bringing potential safety hazards. HUD projects information on the car's front windshield, reducing the risk of the driver's sight deviation caused by bowing his head while reducing driving load and improving driving safety experience [3]. This article adopts the case analysis method, based on the classic design case of navigation head-up display, and analyzes and studies from the aspects of information organization criteria, information content, information structure and visual design, and combines design practice on this basis to initially verify the feasibility of the research, Laid a particular theoretical foundation for the subsequent research on the design of the automotive interactive visual display.

2 Related Work

About 80% of the driver's information is obtained through the visual channel. Therefore, the design of the vehicle interactive visual information display interface is highly problematic [4]. The perspective of information interaction includes a natural display interface for front road conditions, a display interface for assisted driving, an interface for information interaction and entertainment inside and outside the car, and an integrated interface between mobile devices and cars [5]. In the traditional car interior, the primary and auxiliary instrument panels and the central control area are the realization carriers of the information display interface, the so-called head-down display (HDD). However, with the massive involvement of the Internet of Vehicles and information, the display position and display carrier of the car's information will also change accordingly [6].

In the future, smart cars' display will no longer be limited to areas such as traditional dashboards, centre consoles, and rearview mirrors. Any physical equipment and environment may be embedded in the display device and become a medium for information display [7]. The original driving-centric information architecture will be broken down according to entertainment, interaction, and personalized settings, and displayed in different areas of the car.

In terms of display mode, smart cars' display will no longer be limited to a single physical device, but the integration of multiple locations and multiple display forms. From the perspective of human-computer interaction, the most significant advantage of head-up display is that the driver can obtain all kinds of information without leaving the road ahead, effectively improving driving safety, and is considered the safest display method the future one. Smart cars are transitioning to unmanned driving, and driving behaviour will still be vital in automotive human-computer interaction design in the next few years [8]. Therefore, the head-up display has the advantages of improving driving safety and reducing the driver's cognitive load to have a more extensive application scene and space. Even if fully automatic driving is realized in the future, users still have many demands for obtaining information through the front windshield, and the head-up display also has a vast design space [9].

It can be seen from the design research on production cars and concept cars that a significant trend is that the HUD interface will become an essential part of the car's

interactive interface. Recent trends indicate that HUD displays mainly essential driving information (such as vehicle speed, navigation information, etc.) directly related to assisted driving. On this basis, a small amount of unimportant information (such as music in the entertainment system, etc.) can be displayed, but it is in an unimportant area. The future trend is likely to be that HUD, virtual reality, and the 3D display will be combined with the dashboard and even the centre console to form an integrated display interface [10]. The display content will integrate car control, navigation, entertainment, communication and network applications, and future car systems Will form a rugged network terminal. Therefore, this article is based on the classic case of car navigation HUD, analyzes it from the perspective of information organization and visual design, and validates it with design practice.

3 Information Structure of the HUD Interface of the Car Navigation System

The application of HUD in the in-car display system can be traced back to the 1980s, displaying vehicle speed and fuel level with monochrome numbers and symbols. Driven by imaging materials and technology, HUD's display function is also becoming more and more perfect. Taking the navigation system as an example, BMW of Germany and Pioneer of Japan are pioneers in successfully applying navigation system to HUD display. Both are in the navigation product information. The organization, visual design and other aspects all reflect their design characteristics.

3.1 Information Organization Guidelines

The so-called information organization refers to the organization and definition of screen layout, workflow, and interactive behaviour from the interaction framework level. It aims to define product functions, data elements, and the definition of product function modules and operational levels. The navigation system provides users with an information-rich experience based on spatial cognition. Therefore, the driver's recognition and understanding of navigation complex information is the core of HUD information organization guidelines [11]. From the perspective of recognition, attention should be paid to the consistency of the interface layout. The driver can maintain a consistent sense of direction and association in different situations to understand the corresponding association with less cognitive cycles.

For example, the Pioneer HUD display interface is divided into a stable information display area and an iterative information display area. The display content and location of the stable information area are relatively stable, such as place name, direction, time, etc.; while the iterative information area displays different information as the driving situation changes [12]. The content is the real-time push of external driving environment information. For example, when approaching an intersection 50 m, the pre-selected lane reminder icon will be pushed in the iteration area, and it will disappear after passing. The block layout of stable information and iterative information allows users to more quickly and accurately identify changes in the external situation and make driving decisions [13]. From a comprehensible point of view, visual elements familiar to drivers should

be used to organize information to form cognitive migration. For example, the speed limit reminder sign on the BMW HUD navigation interface is the same as the speed limit sign design on real road conditions; the number representing the distance and the arrow indicating the direction are laid out in the same horizontal dimension, which is very similar to the actual road sign design.

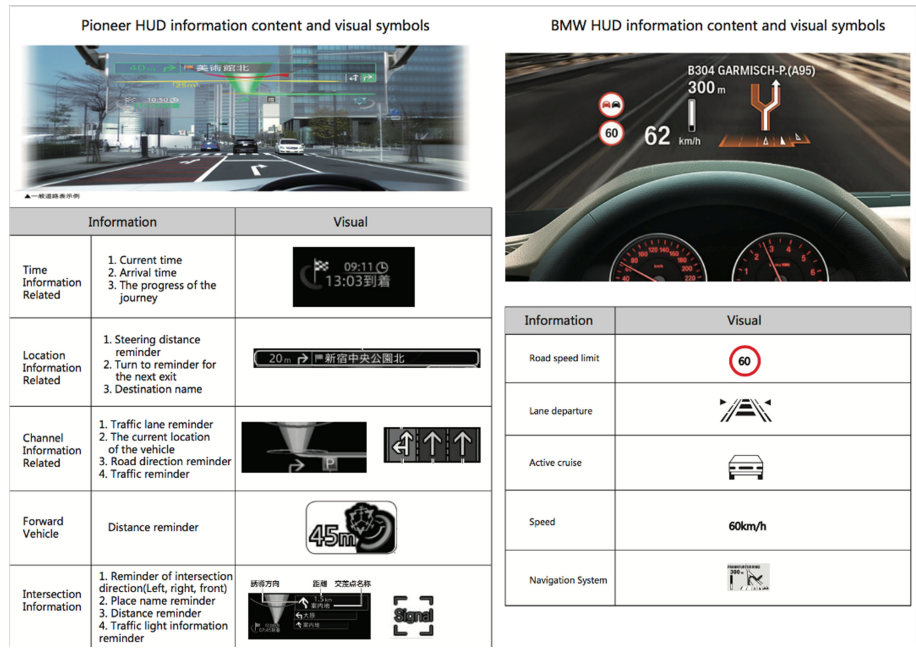


Fig. 1. Pioneer (upper left) from <https://jpn.pioneer/ja/>. BMW (upper right) from <https://www.bmw.com/en/>. HUD information content and visual symbols analyzed by the authors

3.2 Information Content

From the perspective of information content, including functions and data elements. Traditional car navigation product information data elements include maps, numbers, text, graphics, and voice. The data elements presented by the HUD navigation system are mainly numbers, icons, text, and characters, which are represented by visual symbols in the interface [14]. Functional elements are the conversion of functional requirements and the manipulation of data elements. The core function of the navigation system is destination guidance, which can be summarized into location information related (place name, distance reminder), channel information related (direction, lane selection), and assisted driving information related (vehicle speed, progress information reminder, etc.) on the HUD interface. Take the information content of Pioneer and BMW HUD as examples to analyze functions and data elements.

The Pioneer HUD's functional elements shown on the left side of Fig. 1 cover five categories, and each category is further subdivided into functions, namely: time information, location information, channel information, vehicle locking, and intersection information. Taking time information as an example, the functional elements can be subdivided into three items: current time, arrival time, and progress preview. The data elements used in the interface are numbers, characters, and icons. Each functional element corresponds to a combination of data elements. For example, the combination of numbers and clock icons conveys the functional meaning of the current time; the combination of numbers and text conveys the functional meaning of arrival time; the flags and circular track icons convey destination and time progress's functional meaning. The BMW HUD's functional elements include place name, distance, direction, navigation lane, speed, speed limit, and assisted driving. Each function corresponds to the corresponding data element. For example, the speed reminder corresponds to a number, and the direction reminder corresponds to an icon, as shown in Fig. 1 on the right. It can be seen that Pioneer and BMW HUD have certain commonalities in the definition of data and practical information, such as distance, direction, and place name reminders. These elements constitute the core functions of navigation HUD.

3.3 Information Architecture

The purpose of information architecture is to organize several information to quickly and accurately find the information they need. From the point of view of cognitive economy, summarizing and processing diverse visual information through modularization can save drivers' limited cognitive resources, and at the same time, it can make the processing of visual information more timely [15]. Take the Pioneer HUD interface as an example. The information level of the system is wide and shallow. The necessary information is distributed in the interface in the form of modules. The information modules are divided into time information, location information, channel information, etc., and these modules are subdivided into different Functional elements, for example, the location information module contains distance, direction, and destination reminders.

In contrast, the channel information module contains functional elements such as traffic lights, real-time road conditions, and distance reminders of the preceding vehicle, as shown on the left side of Fig. 2. The BMW's HUD display interface organizes information in an "immersive" page. The information module can be divided into two modules: the navigation system and driving assistance. The navigation system module comprises functional elements such as destination name, distance reminder, route reminder, and progress preview. In contrast, the auxiliary driving module comprises practical information such as lane-keeping, active cruise, and speed limit reminder, as shown on the right side of Fig. 2. From the analysis of these two cases, it can be seen that the information module is the integration of information content, and the division of information levels is the basis for the layout of the information module in the interface.

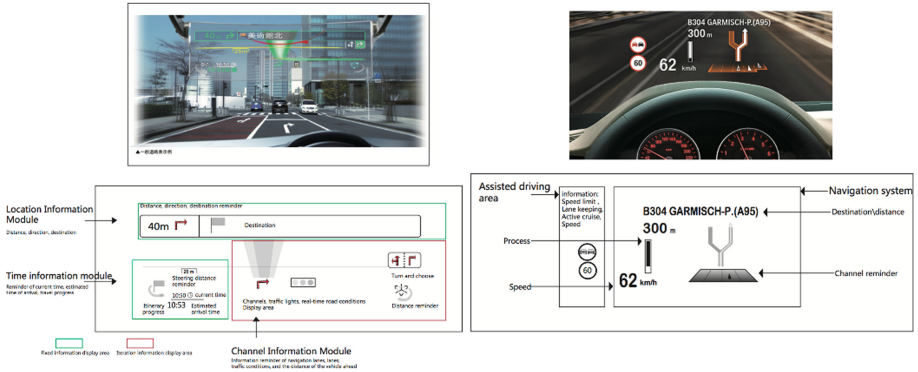


Fig. 2. Pioneer (upper left) from <https://jpn.pioneer/ja/>. BMW (upper right) from <https://www.bmw.com/en/>. HUD information architecture analyzed by the authors

4 Visual Design Analysis of the HUD of the Car Navigation System

Overall, the HUD interface should follow the general rules of visual design, such as interface clarity, readability, style consistency, avoiding visual noise, and having a sense of aesthetic pleasure [14]. In the design details, attention should be paid to HUD's media properties near the HUD imaging materials and lighting environment. For example, the early projection technology was based on a transparent high-refractive-index coating on the front windshield, and the thickness of the film was precisely the peak wavelength of the green selective reflection, which is why the early information display was mostly green. Shana Smith et al. conducted research based on automobile HUD images and driver's perceptual engineering and proposed that colour, font, location and other elements are the key to HUD visual graphic design [15]. Through analyzing the visual interface of BMW and Pioneer HUD, it can be seen that fonts, linearity, auxiliary visual symbols, colours, and auxiliary colour blocks are the design elements that constitute the visual interface of the interface. Among them, colour, text, and icons are the key elements of the HUD visual interface design, as shown in Fig. 3. Also, the HUD interface, the main instrument panel, the auxiliary instrument panel, and the central control interface together constitute the vehicle interactive visual information display interface, so the visual style also needs to be consistent. Taking the interface icons as an example, they are divided into three categories: arrows indicating directions, graphics indicating states (such as time, travel progress), and symbols indicating assisted driving functions (such as lane departure) [16]. These icons are displayed on the HUD interface and the central control and main instrument interface. Therefore, the difference between the HUD display medium and other display media should be considered in the design [3]. The central control display interface is an LCD screen with higher resolution and more prosperous and more delicate colour levels of the icons. HUD is a projection-based windshield display interface with low colour reproduction. This requires designers to determine the hue, brightness, and brightness of the same icon. Saturation needs to be treated differently [17].

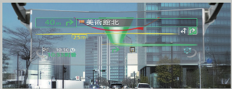
Interface visual		
Font	Fonts: Arial Unicode MS 106km/h	Fonts: Japanese Numeral: 10:50 40m
Line type	The dashed line and the solid line are combined, the solid line indicates the direction and the outline of the road, and the dashed line shows auxiliary information and lane boundaries. 	Thin solid lines and thin dashed lines play the role of information segmentation, and two-way fade lines are used to emphasize the appearance of floating information. 
Visual symbol	 The arrow indicates the driving direction  Triangles for emphasis and guidance	 Arrow indicates pre-alert steering information  Use rectangles to aggregate information
Color	White shows important information: such as place name, vehicle speed, direction; orange and yellow indicate channel information, such as channel departure warning, green indicates speed limit reminder. 	Green displays important information, which is related to driving direction, such as steering reminder, lane selection, distance reminder; white indicates secondary information. 
Auxiliary color block	Express the sense of space through the gradient of hue and transparency, increase the artistic effect of the interface. 	Express the extension of the road through the same color gradient. 
Other	 Overlay color blocks to give the interface a sense of hierarchy	 Superimpose color blocks and increase transparency.

Fig. 3. Pioneer and BMW HUD visual design analysis (Color figure online)

5 Design Practice of Head-Up Display (HUD) of the Car Navigation System

The design strategies are summarized from information organization design and visual design and verified by design practice through the above case analysis.

Follow HUD's information organization guidelines, design with recognition and understanding as to the core. According to the information organization and visual design strategy, the design process can be divided into two steps: 1. Information organization; 2. Visual design.

5.1 Information Organization

First, define the information content, that is, determine the functional elements like direction, distance, place name, navigation lane, pre-selected lane, and speed limit reminder. According to the division of HUD information levels, the above functional elements are summarized into primary channel information related and auxiliary information related to the hierarchical relationship. The purpose is to provide a basis for the spatial layout of information modules.

Second, follow the design guidelines of consistency, reduce cognitive cycles, and structure information according to the information content level, as shown in Fig. 4.

The interface is divided into two modules: the main channel information module and the auxiliary information module. The functional elements contained in the main

channel information module are of great significance to the user’s driving decision-making. Therefore, the layout is in the centre of the interface and coincides with the vision’s main driving field. The structure is stable and consists of direction, distance, place name, and channel reminders. The auxiliary information module presents iterative information, which is only displayed in the context that needs to be displayed, so the layout is on the upper right of the interface to avoid interference with the A-pillar. For example, the current speed and speed limit reminder icon will only appear in a speeding situation.

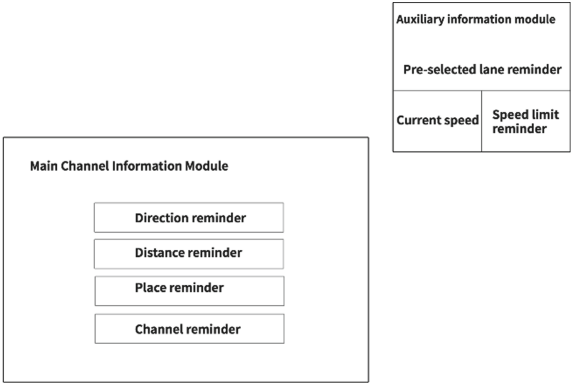


Fig. 4. Information architecture

5.2 Visual Design, that is, Visual Symbol and Interface Design

According to the above-defined functional elements, visual symbols are designed, and the data elements used are a combination of numbers, characters and icons. Follow the comprehensibility principle, and indicate functions such as lane reminders and speed limit reminders based on road signs familiar to drivers, as shown in Fig. 5. In this case, the HUD display position is in the driver’s primary field of view. To reduce interference, follow the visual design strategy and repeatedly scrutinize the size and colour of arrow symbols, numbers, and characters. For example, when dealing with visual symbols’ proportional relationship, The division of information levels is the basis, and the direction reminder function is the primary information, as shown in Fig. 6. Therefore, the arrow symbol takes a larger proportion in the main channel information module than the distance and place name reminder icons. When dealing with the relationship between the channel reminder and the road surface change, the method adopted expresses the road extension distance and space relationship through three colour band symbols with gradual transparency, as shown in Fig. 7.

Information		Vision
Main channel information related	Direction reminder	
	Distance reminder	50 m
	Place name reminder	Hyde Park
	Channel reminder	
Auxiliary information related		  

Fig. 5. Visual symbol

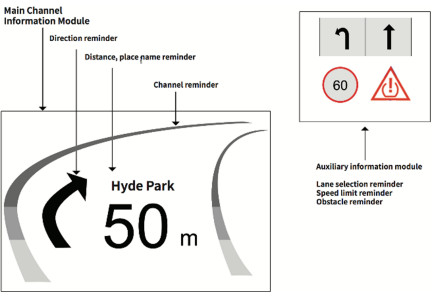


Fig. 6. V interface low-fidelity prototype

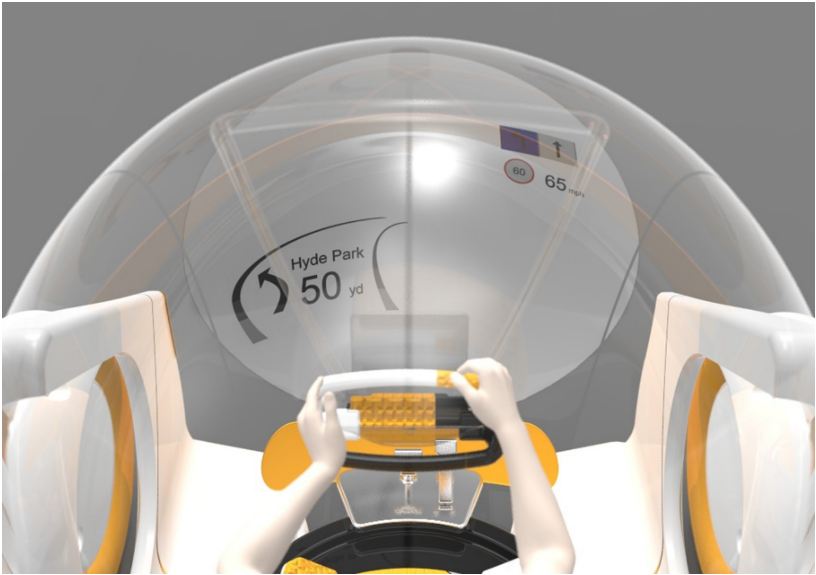


Fig. 7. Interface high-fidelity prototype

6 Conclusion

The use of HUD as the navigation system's display interface has mature applications in the aviation field. However, the HUD as a car navigation interface is still in the stage of continuous exploration and improvement, which is also the practical significance of this paper. This paper analyzes Pioneer and BMW's HUD navigation system, proposes the HUD interface information organization strategy and visual design strategy, and has been initially verified through design practice. This research provides an essential theoretical,

and practical basis for designing an automotive interactive visual information display interface.

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